

PROJECT REPORT

ON

JAVA WEBSITE GENERATOR

A Java Swing-Based Application for Automatic Website Generation

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Project Overview

The Java Website Generator is a GUI-based application developed using Java Swing that automatically creates basic websites based on user input. Users can select a website type, enter details, and generate an HTML webpage instantly. The project demonstrates Java GUI development, file handling, event handling, and dynamic HTML generation in a simple and user-friendly manner.

Technologies Used

- Java
- Java Swing
- HTML
- File Handling
- Event Handling
- Exception Handling

"Automating Website Creation Through Java and HTML"

Introduction

The Java Website Generator is a desktop-based application developed using Java Swing that automatically creates basic websites based on user input. The application provides an easy-to-use graphical interface where users can select a website type, enter relevant details, and generate a complete HTML webpage without manually writing HTML code.

This project demonstrates the integration of Java GUI development, event handling, file handling, and web technologies. It is designed for beginners who want to create simple websites quickly and efficiently.

Objective

- To simplify the process of website creation.
 - To generate HTML webpages automatically using user input.
 - To provide a user-friendly graphical interface for website generation.
 - To demonstrate the use of Java Swing, file handling, and event handling.
 - To reduce the effort required for creating basic websites manually.
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Problem Statement

Many beginners find it difficult to create websites because they need knowledge of HTML and web development concepts. Writing website code manually can be time-consuming and confusing for new learners.

The Java Website Generator solves this problem by allowing users to generate basic websites through a simple graphical interface without requiring extensive knowledge of HTML coding.

Technologies Used

Programming Language

- Java

GUI Framework

- Java Swing

Web Technology

- HTML

Concepts Used

- Object-Oriented Programming (OOP)
- Event Handling
- File Handling
- Exception Handling
- GUI Development

Java Classes Used

- JFrame
 - JLabel
 - JTextField
 - JTextArea
 - JButton
 - JComboBox
 - FileWriter
 - Desktop
-

Features

- User-friendly graphical interface.
 - Generates websites automatically.
 - Multiple website templates available.
 - Restaurant website generation.
 - Portfolio website generation.
 - Business website generation.
 - Dynamic HTML content creation.
 - Automatic file creation and storage.
 - Opens generated website directly in browser.
 - Error handling using try-catch blocks.
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Working of the Project

1. The user launches the application.
2. The application displays a graphical interface.
3. The user selects a website type:
 - Restaurant
 - Portfolio
 - Business
4. The user enters the website name.
5. The user enters details about the website.
6. The user clicks the "Generate Website" button.

7. The application generates HTML code dynamically.
 8. The HTML code is written into an index.html file.
 9. The generated website is automatically opened in the default browser.
 10. A success message is displayed to the user.
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Program

```
import javax.swing.*;
import java.awt.*;
import java.io.*;

public class WebsiteGeneratorApp {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Website Generator");
        frame.setSize(500, 480);
        frame.setLayout(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        frame.getContentPane().setBackground(new Color(167, 199, 231));

        JLabel l1 = new JLabel("Website Type:");
        l1.setBounds(20, 20, 120, 25);

        String[] types = {"Restaurant", "Portfolio", "Business"};
        JComboBox<String> box = new JComboBox<>(types);
        box.setBounds(160, 20, 250, 25);

        JLabel l2 = new JLabel("Name:");
        l2.setBounds(20, 60, 120, 25);

        JTextField nameField = new JTextField();
        nameField.setBounds(160, 60, 250, 25);

        JLabel l3 = new JLabel("Details:");
        l3.setBounds(20, 100, 120, 25);

        JTextArea details = new JTextArea();
        details.setBounds(160, 100, 250, 80);

        JButton btn = new JButton("Generate Website");
        btn.setBounds(150, 210, 180, 35);

        JLabel status = new JLabel("");
        status.setBounds(20, 260, 430, 25);

        btn.addActionListener(e -> {
```



```

// Business Website
else {

    html =
        "<html><head><title>" + name + "</title></head>" +
        "<body style='font-family:Arial;background:#e8f5e9;padding:20px;'>" +
        "<div style='background:white;padding:20px;border-radius:15px;'>" +
        "<h1>" + name + "</h1>" +
        "<img src='profile.jpg' width='300'>" +
        "<p>" + info + "</p>" +
        "<h2>Services</h2>" +
        "<ul>" +
        "<li>Consulting</li>" +
        "<li>Software Development</li>" +
        "<li>Technical Support</li>" +
        "</ul>" +
        "<h2>Contact</h2>" +
        "<p>Phone: 9XXXXXXXXXX</p>" +
        "</div></body></html>";

}

FileWriter fw = new FileWriter("index.html");
fw.write(html);
fw.close();

Desktop.getDesktop().browse(new File("index.html").toURI());

status.setText("Website created successfully!");

} catch (Exception ex) {

    status.setText("Error creating website.");
}
});

frame.add(l1);
frame.add(box);

frame.add(l2);
frame.add(nameField);

frame.add(l3);
frame.add(details);

frame.add(btn);
frame.add(status);

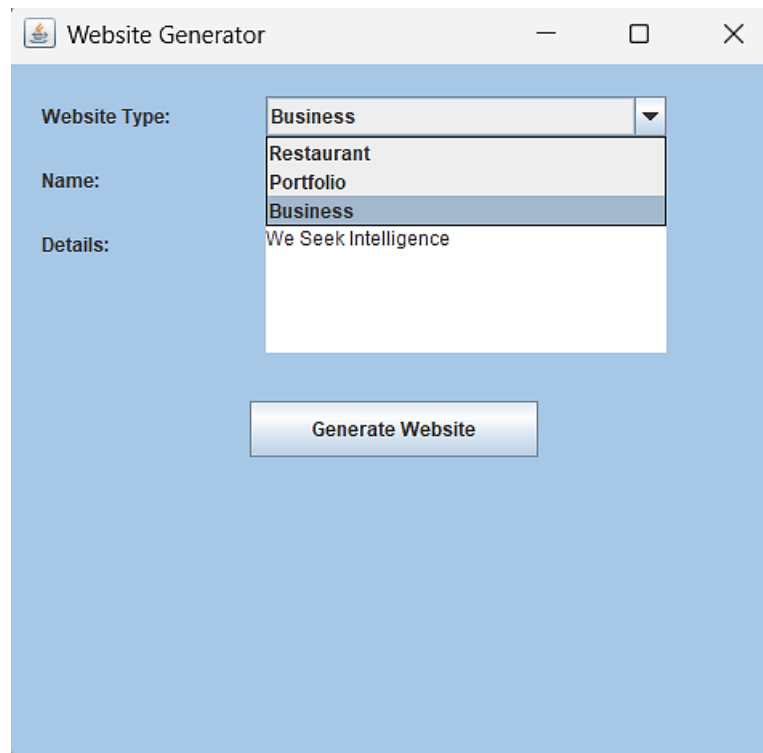
frame.setVisible(true);
}

```

}

Output

Step I: Select the website type



The screenshot shows a window titled "Website Generator" with a light blue background. On the left, there are three labels: "Website Type:", "Name:", and "Details:". To the right of "Website Type:" is a dropdown menu with a list of options: "Business", "Restaurant", "Portfolio", and "Business". The first "Business" option is highlighted. Below the dropdown menu is a text input field containing the text "We Seek Intelligence". At the bottom center of the window is a button labeled "Generate Website".

Step II: Enter the name and details and then click on generate website

Website Generator

Website Type:

Portfolio

Name:

Jyoti Restaurant

Details:

We Serve Excellent Quality of Food!

Generate Website

Website created successfully!

RESTAURANT WEBSITE

Jyoti Restaurant



About Us

We serve excellent quality of food!

Menu

Dish	Price
Paneer Butter Masala	220
Veg Biryani	180
Butter Naan	40

Address

Pune, India

PROFILE WEBSITE

Gayatri Sable



Hobbies : Reading, Webprogramming

Skills

- Java
- Web Development
- Artificial Intelligence

Contact

Email: example@gmail.com

BUSINESS WEBSITE

Nova Tech



We seek Intelligence!

Services

- Consulting
- Software Development
- Technical Support

Contact

Phone: 9XXXXXXX

Advantages

- Easy to use.
- No manual HTML coding required.
- Saves time in website creation.
- Beginner-friendly application.
- Interactive graphical interface.
- Supports multiple website templates.
- Demonstrates practical Java concepts.
- Can be extended with additional features.

Future Scope

- Add more website templates.
- Allow image upload functionality.

- Add CSS theme customization.
- Support multiple web pages.
- Generate responsive websites.
- Add database connectivity.
- Export websites as ZIP files.
- Add drag-and-drop design features.
- Include user authentication system.
- Support additional web technologies such as CSS and JavaScript generation.

Conclusion

The Java Website Generator is a useful application that simplifies website creation through automation. The project successfully demonstrates Java Swing GUI development, file handling, event handling, exception handling, and dynamic HTML generation. It provides an efficient solution for generating basic websites and serves as an excellent learning project for understanding software development concepts.